

DoT: Degradation of Throughput by the Adversarial Behavior of Nodes in Terahertz Communication Networks

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Abstract

The inherent high directivity nature of the THz signals requires the receiver and transmitter to be face-to-face in order to communicate successfully. To address this challenge, several medium access control (MAC) protocols have been proposed for link-level synchronization based on receiver-initiated handshake. Receiver-initiated handshake-based protocols, although they guarantee synchronization, pose a vulnerability that can be exploited by an adversary to degrade the network throughput. In this work, we study how the overall throughput is impacted when some adversarial nodes participate in the receiver-initiated handshake but do not complete the communication. Our experimental results through ns-3 simulations show that throughput is drastically reduced in our proposed *Degradation of Throughput (DoT)* scenario when more and more nodes act as adversaries.

CCS Concepts

• **Networks** → **Link-layer protocols**; **Network simulations**; *Wireless access networks*.

Keywords

Terahertz communications, 6G, ns-3, Throughput analysis

ACM Reference Format:

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1 Introduction

While the deployment of 5G is being done, people are also searching for Beyond 5G (B5G) for enhanced network performance. Moving the operations into the THz band (0.1 THz to 10 THz) can meet the growing needs in the future [4]. This region in the electromagnetic spectrum is considered to be the least explored region till now [18]. Although considering the THz band to achieve high data rates is not a very new idea, the lack of physical devices that can function properly at these frequencies was a hindrance to the improvement.

However, many recent advancements in device technologies [15] have shown us the light to overcome that.

The peculiar characteristics of the THz band make it different from low-frequency bands. The propagation of the THz signal is mainly affected by the absorption of the molecules in the transmission medium. Due to the small wavelength, the power of a THz signal is greatly affected by the molecular absorption as it spreads through the medium. If the communication distance is below one meter, molecular absorption defines a single transmission window, whereas for a distance above one meter, multiple transmission windows are opened. These two scenarios are considered nanoscale and macroscale scenarios, respectively [18].

To overcome the spreading loss, THz links rely on highly directional beams generated from highly directional antennas [14] in both the transmission and reception ends. This characteristic makes THz communication different from conventional communication. To overcome the challenges for THz communication, various MAC protocols have been proposed in recent years. Some of the relevant works have been mentioned in a survey [5].

The use of directional antennas (DA) creates another challenge for THz communication. Nodes can communicate only when they are aligned with the access point (AP). In order to tackle this deafness problem, rotating antennas are used, which makes the link-level synchronization challenging. To overcome this challenge, some MAC layer protocols have been proposed that rely on a receiver-initiated handshake. ADAPT-3 [11] is one of the receiver-initiated handshake-based MAC protocols that completes a 3-way handshake between transmitter and receiver before making the data transmission.

In this work, we present a novel adversarial scenario, *Degradation of Throughput (DoT)*, for the receiver-initiated 3-way handshake-based protocols for Terahertz communications. In the proposed scenario, some nodes deliberately respond to the first control packet of the receiver but do not send any data packets, and thus do not complete the communication. For the experiment, we used ns-3 [12] based THz simulator *TeraSim* [6]. Our simulation results show that the throughput of the overall network is greatly reduced when there are more nodes playing this adversarial role. The contributions of the paper are as follows:

- We introduce an adversarial scenario for receiver-initiated 3-way handshake-based MAC protocol for THz communication networks.
- We present our experimental results from ns-3 simulations, which validate that adversary nodes can significantly degrade network throughput by exploiting the proposed scenario.



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The rest of the paper is organized as follows. In Section 2, we discuss some of the recent studies on Terahertz MAC protocols. In Section 3, we describe the ADPAT-3 MAC protocol. We present our adversarial scenario in Section 4. In Section 5, we describe the simulator settings, network topology, and also present our findings. We show the effect of having *white list*, a feature of the ADPAT-3 MAC protocol, enabled in Section 6. Later in Section 7, we provide an argumentative analysis of the difference between Degradation of Throughput (DoT) and Denial-of-Service (DoS), and also a comparative analysis of DoT with other existing works. Finally, we conclude this work in Section 8.

2 Related Work

Authors in [18] proposed a receiver-initiated handshake-based MAC protocol for the macroscale scenario of THz. The receiving node rotates at high speed and broadcasts a Clear-To-Send (CTS) packet while rotating. The transmitter receives the CTS packet when it gets aligned face-to-face with the receiver. The transmitter sends a DATA packet, if it has any to send, after waiting for a random backoff time. After the successful transmission of a DATA packet, the receiver sends an ACK packet to the transmitter, which contains a flag indicating whether the CTS is still valid for the next DATA transmission. An enhanced version of this protocol, ADAPT-3, is presented in [11] for the macroscale scenario. ADAPT-3 protocol relies on receiver-initiated 3-way handshake for link-level synchronization. Although the exchange of three control packets introduces overhead, the results presented in [11] show that more than 100 Gbps throughput can be achieved.

In recent years, ADAPT has drawn significant attention to the research community, and many recent studies have focused on the applicability of the ADAPT protocol for different use cases. For example, Absaruddin et al., [2] studied the impact of antenna beamwidth on the performance of the ADAPT protocol in the context of wireless data centers. Their analysis confirmed the tradeoff between antenna beamwidth and different performance metrics like throughput and discovery time. Considering this tradeoff, the same authors in their other work [3] proposed a new MAC protocol for wireless data centers named DBS-ADAPT, on top of the ADAPT-3 protocol. In DBS-ADAPT [3], client nodes and the access point utilize the narrower beamwidth during control packet transmission. The access point calculates a wider beamwidth based on the client node's position received during the control packet transmission. During the data transmission phase, client nodes and the access point select this wider beamwidth.

Korkmaz et al., [7] also studied the dynamic beamwidth adjustments of the ADAPT-3 protocol, focusing on drone-to-drone communication scenarios. Similar to the DBS-ADAPT [3], the nodes (drones) and access points here also share location information during the transmission of control packets and adjust their beamwidth accordingly. Data transmission happens with the adjusted beamwidth. After the completion of data transmission, both the client node and the access point return to their initial beamwidth, which is determined based on the drone's capabilities.

3 Background

In this section, we present an overview of the ADAPT-3 MAC protocol that was proposed for the THz macroscale scenario.

Morales et al., [11] proposed ADAPT, a receiver-initiated handshake-based MAC protocol for THz macroscale scenarios. This protocol assumes that the AP has a turning antenna that periodically rotates. Here, the coverage area of the access point (AP) is divided into some virtual sectors. In a particular sector, the AP broadcasts a Call-To-Action (CTA) packet, which indicates its availability to begin a communication, and waits for a fixed time interval. During this wait time, if a client node receives the CTA packet and has data to send, it responds with a Request-To-Send (RTS) packet. Upon receiving an RTS packet, the AP sends another control packet, Clear-To-Send (CTS), and grants access to the channel to the client node. The client node then sends the data. After successfully receiving the DATA packet, the AP sends an ACK packet.

During the wait time in each sector, if the AP receives multiple RTS packets, it responds with CTS packets to all nodes. Each of the nodes receives a transmission slot during which it needs to transmit the data in order to avoid a collision. The AP, after completing the data transmission for all the nodes, moves to the next sector. If no RTS is received in the initial wait time, AP moves to the next sector and completes the same procedure again.

Authors also introduced an adaptive Modulation and Coding Scheme (MCS) mechanism for the 3-way handshake. During the control packet transmission, based on the received signal power, the AP decides which MCS should be utilized by the node.

The AP can also maintain a white list for each sector, a mapping of each node to a single sector. The AP does not respond to an RTS packet if that is sent from a node that does not belong to that particular sector's white list. This ensures that nodes closer to the AP cannot utilize adjacent sectors to communicate.

4 Methodology of DoT

In ADAPT-3 [11], after sending the CTS packet, the AP waits for a fixed time for the client node to complete the data transmission. Whether the data transmission is successful or not, the AP will wait during that specific time. This property can be misused by an adversary node in the network.

In our proposed DoT scenario, an adversary node exploits the fact that the AP, after completing the 3-way handshake, waits for a fixed time. Here, the adversary node sends the RTS packet when it receives a CTA packet from the AP, even if there is no data to be sent. The adversary node takes channel access by completing the 3-way handshake, but it does not send any data afterwards.

We present our proposed adversarial scenario, DoT, in a diagram in Fig. 1. Here, we consider that there is an adversarial node in a sector to which the AP is directed. The access point sends the CTS packet, and the adversary node responds with an RTS packet. The AP waits for a fixed time to receive any other RTS. After that, it sends the CTS packet and grants access to the channel to the node. The node, upon receiving the CTS packet, is supposed to send a DATA packet. But here it acts as an adversary. Even after receiving the access, the adversary node does nothing. The AP in this case has to wait for a fixed sector time. After the sector time finishes, it moves to the next sector. The AP will come again to

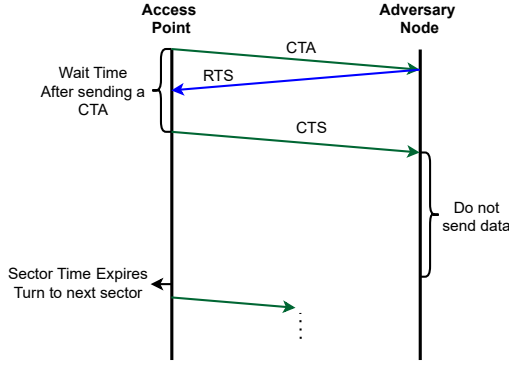


Figure 1: Packet flow between the AP and an adversary node

this sector in the next round. The adversary node will continue doing its adversarial play. In this way, the adversarial node is able to block the AP, but no data transmission happens. Thus, the overall throughput of the network is affected because the AP is unable to utilize this time to communicate with a valid node.

If there are multiple adversary nodes in a single sector, the AP will grant access to all of the adversarial nodes in turn. Thus, all the adversarial nodes can collectively block the AP for a longer time since the AP will now wait longer, expecting that multiple data transmissions may happen, before moving to the next sector.

5 Experimental Evaluation

In this section, we present the details about our experimental analysis. We first present the configurations that we used for the simulator. Then we present an example scenario of the node positions that we used in our simulation. Later, we describe our findings.

5.1 Simulator Configuration

For our experimentation, we used the network simulator *ns-3*. Specifically, we used TeraSim [16, 17], which is an *ns-3* module tailored for THz network simulations. The physical layer configuration of TeraSim follows the IEEE 802.15.3d standard [1, 13] for THz band communications.

In our simulation, we disabled the whitelist feature of ADAPT-3. This ensures that the adversary node, when placed closer to the access point, can utilize the adjacent sectors along with its primary sector. We also turned off the adaptive MCS feature of ADAPT-3.

In our experiment, nodes utilize QPSK modulation, which is one of the supported modulations for THz single carrier mode according to the standard [13]. We ran our simulations for a fixed inter-arrival time of 500 μ s. In one configuration, we considered the range of 8.4 m, when the area is divided into 15 virtual sectors. In another configuration, we considered a range of 34 m with 30 sectors.

5.2 Network Topology

In our experiments in *ns-3*, the access point is located in the center. All the other nodes are positioned randomly within the coverage area of the access point.

We first considered the scenario where there is a fixed number of valid nodes and no adversary node. Thus, it reflects the case

when the nodes follow the ADAPT-3 protocol properly. Later on, we add adversary nodes to that network. To better visualize the proportional changes in performance metrics, we keep the number of adversary nodes as a percentage of the total number of valid nodes.

For each simulation setting with a fixed number of valid nodes and adversary nodes, we ran our simulations for five iterations. For each iteration, we changed the location of the valid nodes and adversary nodes and then calculated the performance metrics. The value presented in the result section is the average of the five iterations. In Fig. 2, we present relative node positions for each iteration for a particular case in our experiment. Here, we considered 20 valid nodes and 4 adversary nodes placed within a range of 34 m.

5.3 Results

In our experiment, we consider two performance metrics - *Throughput* and *Average Packet Time*. We show our simulation results as *Throughput* or *Average Packet Time* against the number of adversary nodes in Fig. 3 to Fig. 6. We determine the number of adversary nodes as a percentage of the total valid node count. For example, for 10 valid nodes, 0% adversary nodes means there is no adversary node. 20% adversary nodes means apart from the 10 valid nodes, there are 2 nodes that act as adversary nodes. Similarly, for 20 and 30 valid node counts, 20% adversary node count represents the presence of 4 and 6 adversary nodes, respectively.

Fig. 3 shows the throughput changes with the inclusion of adversary nodes when the access point has a coverage area of 8.4 m range. For 10 valid nodes, when there is no adversary node present in the network, the overall throughput can be up to 91 Gbps. However, the throughput is drastically reduced when there are some adversary nodes present in the network. Even with the inclusion of 20% adversary nodes, the throughput can be reduced by more than 50%. In this case, 2 adversary nodes placed alongside 10 valid nodes reduce the network throughput from around 91 Gbps to 38 Gbps. The throughput continues to drop when there are more adversary nodes. If we deploy the same number of adversary nodes as the valid nodes, the network throughput becomes very low. For example, for 10 valid nodes and 10 adversary nodes, the throughput becomes only around 15 Gbps, whereas it was around 91 Gbps without any adversary nodes. We can see from Fig. 3 that the reduction rate is higher when we increase the total valid node numbers. For example, 20% adversary nodes in the 30 valid nodes scenario can reduce the throughput by more than 70%. With the increase in the number of adversary nodes, the throughput is also decreased gradually.

In Fig. 4, the average packet time for the above configuration of 8.4 m range is shown. When there is no adversarial node, the average packet delay is in the range of 5.91 to 6.94 μ s for our simulations. The nodes that play as adversaries can impact the average packet time for other non-adversary nodes significantly. With 40% adversary nodes, the packet time average is increased up to 7 times for 10 valid nodes, and up to 20 times for 30 valid nodes.

Fig. 5 and Fig. 6 show the simulation results for a 34 m range with 30 virtual sectors. For 10 valid nodes, the throughput is reduced by more than 50% when 40% of the nodes are adversarial. But for 20 valid nodes, only 20% adversarial nodes can reduce the throughput

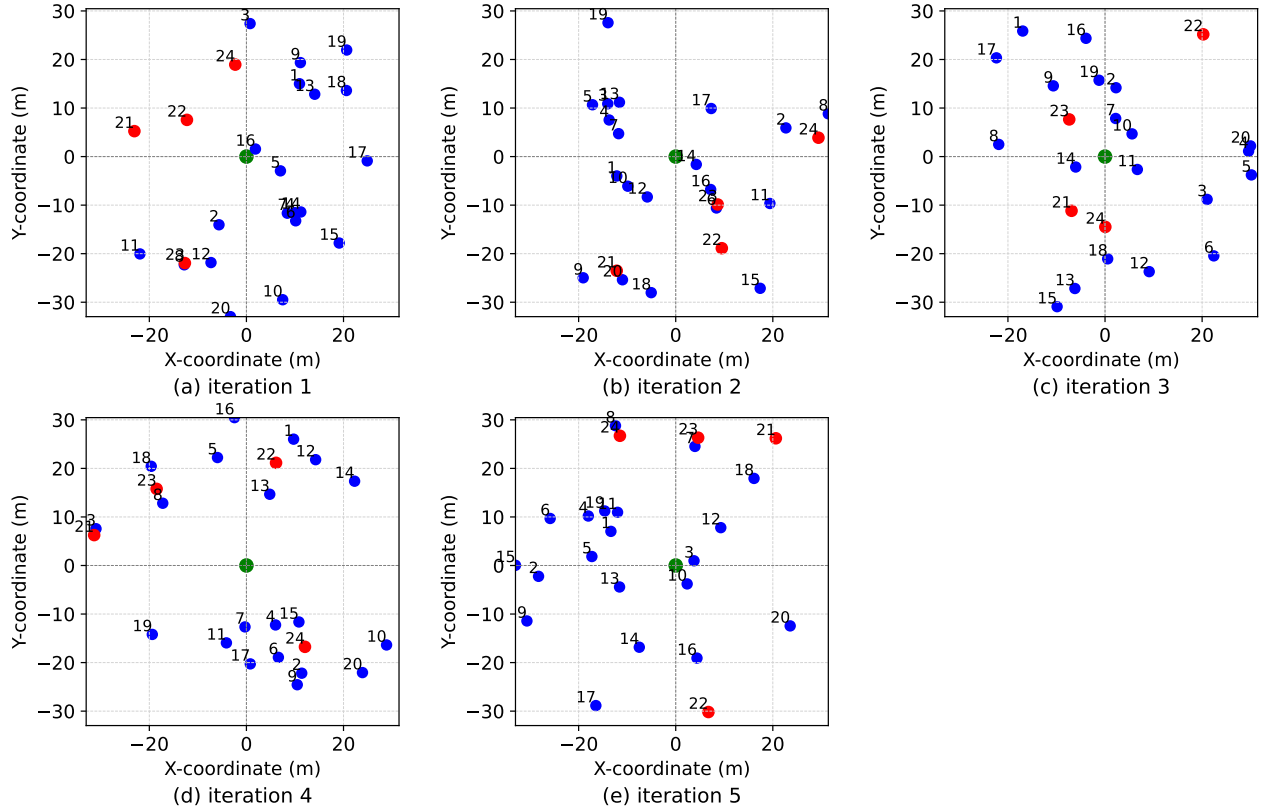


Figure 2: Relative node positions for 20 valid nodes and 4 adversary nodes (for a range of 34 m) [AP in green, valid node in blue, and adversary node in red]

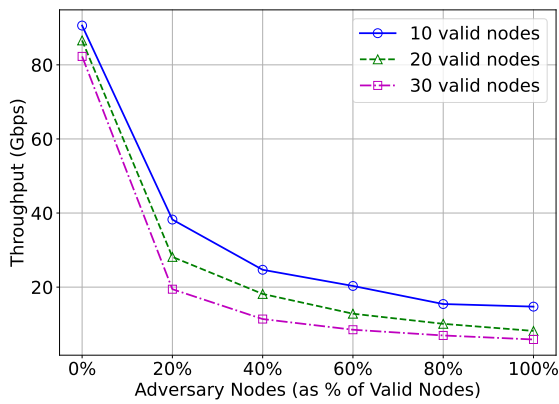


Figure 3: Impact on throughput (for a range of 8.4 m)

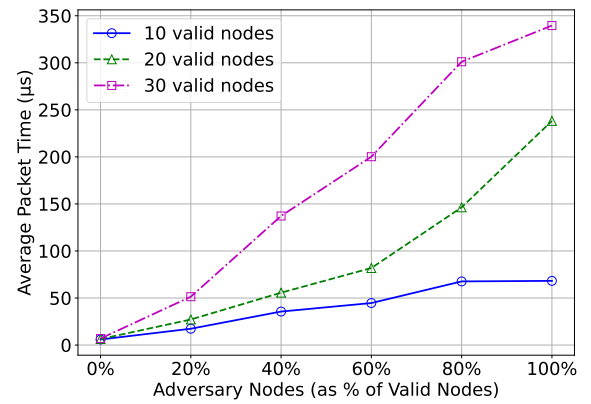


Figure 4: Average packet time (for a range of 8.4 m)

by 50%. This also shows that when the node density increases, the impact of our proposed adversarial scenario also gets higher.

With lower node density, the overall average packet time for adversarial scenarios is also reduced compared to our previous run configuration. However, it also shows a gradual rise with the

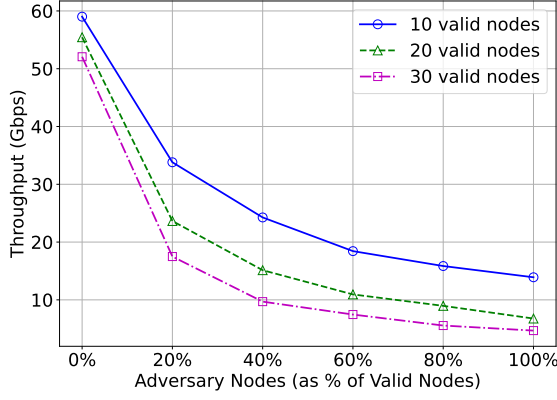


Figure 5: Impact on throughput (for a range of 34 m)

increase of adversary node percentage, which is shown in Fig. 6. Overall, the results obtained from the simulations show that adversarial nodes can degrade the overall throughput significantly.

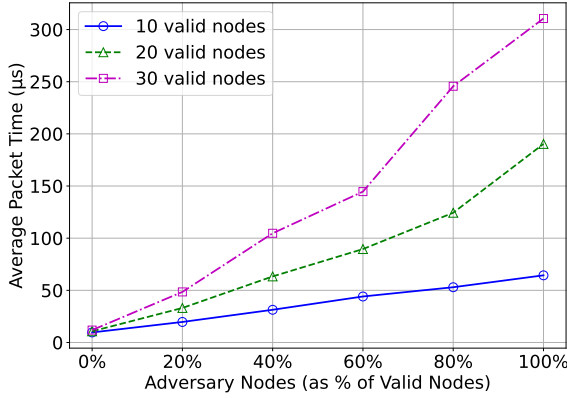


Figure 6: Average packet time (for a range of 34 m)

For each combination of valid nodes and adversary nodes, we ran the simulation five times. For each iteration, the node positions were changed. Thus, the results obtained from each iteration slightly differ from those of the other iterations. Each value in the graphs in Fig. 3 to Fig. 6 is the average value of these five iterations. In Fig. 7 to Fig. 10, we present the average values that we used for the earlier graphs. However, this time we also present the standard deviation as error bars.

From Fig. 7, we can see that throughput values for each iteration for any simulator run configuration did not deviate much from the average value of that corresponding run configuration.

In Fig. 8, we show the standard deviation of the calculated packet time from five iterations for each of the node combinations. Here we see that with the increase in node numbers, the deviation of the packet time values also increases.

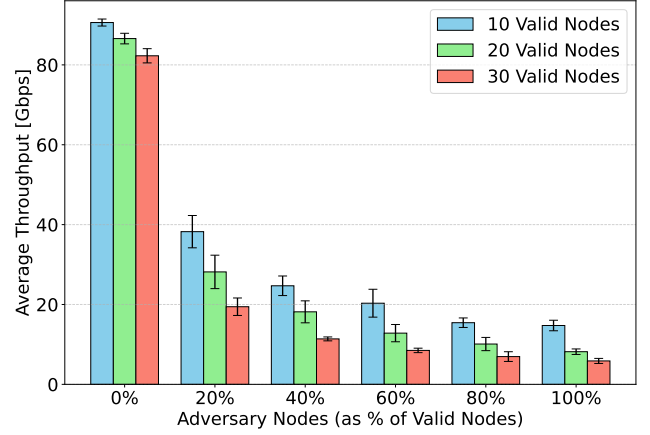


Figure 7: Throughput average from five iterations (8.4 m range)

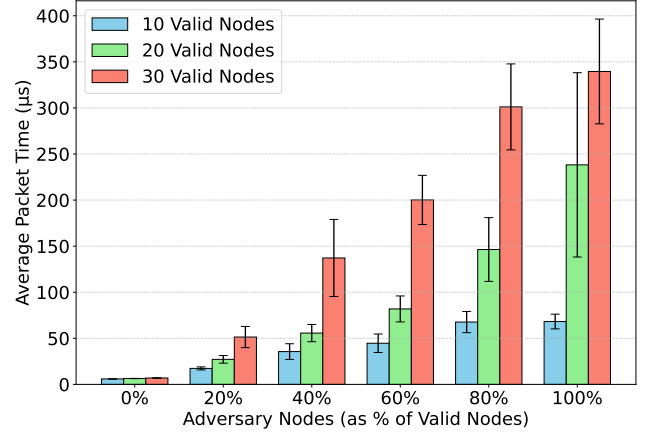


Figure 8: Packet time average from five iterations (8.4 m range)

A similar trend can be observed for the standard deviation of throughput and average packet time for the range of 34 m, which are shown in Fig. 9 and Fig. 10. Throughput values usually do not vary that much for any configuration of the simulation, whereas the average packet time values vary significantly, especially with the increase in node numbers.

We summarize our findings in Table 1 and Table 2. Table 1 represents the throughput changes in our proposed DoT scenario for different numbers of valid node and adversary node combinations. Here, the negative cell value means a reduction in the throughput.

Similarly, in Table 2, we present the percentage by which the adversary nodes can increase the average packet time in the DoT scenario compared to the scenario when there is no adversary.

6 Effect of White Listing

In order to ensure fair distribution of data transmission opportunity per cycle for each node, authors in ADAPT-3 [11] proposed that AP

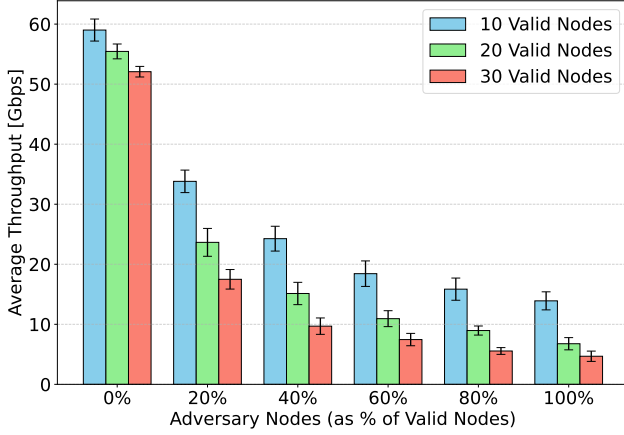


Figure 9: Throughput average from five iterations (34 m range)

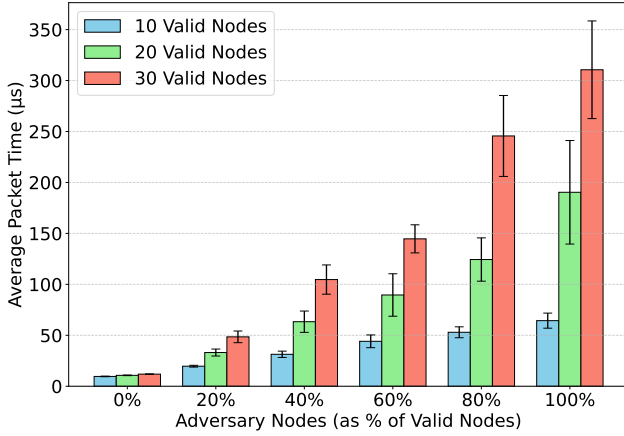


Figure 10: Packet time average from five iterations (34 m range)

Table 1: Percentage change in Throughput for different numbers of adversarial nodes (as % of valid nodes)

Range	Number of valid nodes	Percentage change in Throughput against percentage change in adversary nodes				
		20%	40%	60%	80%	100%
8.4 m	10	-57.80%	-72.76%	-77.57%	-82.96%	-83.74%
	20	-67.48%	-79.03%	-85.19%	-88.34%	-90.56%
	30	-76.37%	-86.17%	-89.67%	-91.55%	-92.87%
34 m	10	-42.70%	-58.88%	-68.76%	-73.13%	-76.43%
	20	-57.35%	-72.71%	-80.27%	-83.84%	-87.80%
	30	-66.40%	-81.39%	-85.68%	-89.33%	-91.01%

can keep a list of nodes per sector. In a particular sector, AP only communicates with nodes that belong to the sector's white list.

In Fig. 11, we present a comparison of throughput degradation for various node combinations when the white list feature is turned on and off. For 10 valid nodes, keeping the white list on slightly

Table 2: Percentage change in Average Packet Time for different numbers of adversarial nodes (as % of valid nodes)

Range	Number of valid nodes	Percentage change in Packet Time against percentage change in adversary nodes				
		20%	40%	60%	80%	100%
8.4 m	10	193.52%	502.30%	655.29%	1044.37%	1054.06%
	20	326.00%	772.66%	1183.81%	2194.77%	3635.09%
	30	640.79%	1877.91%	2785.90%	4241.55%	4795.78%
34 m	10	104.22%	225.45%	357.72%	449.69%	568.08%
	20	208.95%	491.34%	736.15%	1060.99%	1677.06%
	30	307.70%	780.49%	1116.54%	1965.77%	2512.00%

improves throughput compared to when it is off. However, the reduction of network throughput with the inclusion of adversarial nodes still remains significant, which confirms the effectiveness of our adversarial scenario. In the case of 20 and 30 valid nodes, the addition of adversarial nodes decreases the network throughput in almost the same way whether we use the white list feature or not. This confirms that utilizing the white list does not guarantee mitigation against our proposed adversarial scenario, DoT.

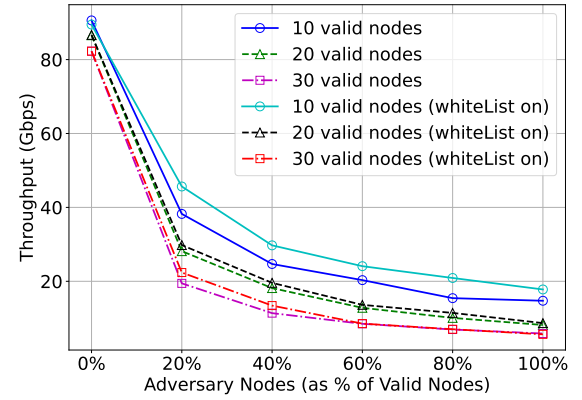


Figure 11: Impact on throughput, with white list on and off (for a range of 8.4 m)

7 Discussion

In this section, we describe how our proposed scenario is different than Denial-of-Service (DoS). Then we provide a comparative analysis of our study with existing works.

7.1 DoT vs DoS

In Denial-of-Service (DoS) attacks, the attacker makes sure that the server is unreachable for legitimate users. However, in our proposed scenario, the access point is reachable through valid nodes. But the protocol design of the receiver-initiated 3-way handshake poses a vulnerability that can be exploited by an adversarial node to make the AP wait unnecessarily for a specific time. In our scenario, the adversarial node does not select the time period for which the AP waits. The nodes can play the adversary simply by not completing the data transmission.

Table 3: Comparison with Related Studies

Research Study	Receiver-initiated handshake	3-way handshake	THz macroscale scenario	Exploitation over MAC protocol	Degradation of Throughput through adversary
Korkmaz et al., 2025 [7]	✓	✓	✓	✗	✗
DBS-ADAPT (Absaruddin et al., 2024) [3]	✓	✓	✓	✗	✗
Absaruddin et al., 2024 [2]	✓	✓	✓	✗	✗
ADAPT (Morales et al., 2021) [11]	✓	✓	✓	✗	✗
Xia et al., 2019 [18]	✓	✗	✓	✗	✗
Xia et al., 2017 [19]	✓	✗	✓	✗	✗
DRIH-MAC (Mohrehkesh et al., 2015) [10]	✓	✗	✗	✗	✗
RIH-MAC (Mohrehkesh et al., 2014) [9]	✓	✗	✗	✗	✗
Mohrehkesh et al., 2014 [8]	✓	✗	✗	✗	✗
DoT (Our proposed)	✓	✓	✓	✓	✓

The valid nodes in the network can still communicate with the AP, but the time for completing a successful communication increases due to the adversarial behavior of client nodes. This adversarial scenario causes the overall network throughput to drop. To summarize, our proposed scenario is different than DoS, and therefore, we coined the term *DoT (Degradation of Throughput)* for it.

7.2 Comparative Analysis

In Table 3, we present some of the related studies of THz MAC protocols and provide a comparison with our work based on some important aspects. We consider those protocol studies that rely on receiver-initiated communication. We took the following features to compare with our work: whether the study considers a 3-way handshake or not, whether it focuses on a macroscale scenario of Terahertz, whether the work discusses any vulnerability exploitation over the proposed MAC protocol, and finally, whether it considers the adversarial scenario that we proposed in our work. We can see from the table that no other work, apart from ours, has considered any exploitation of the MAC protocols. Our work is the first approach to conduct the study on throughput degradation by exploiting the proposed MAC protocol through an adversary.

8 Conclusion

In this work, we present an adversarial scenario for THz communication, which we term as *DoT (Degradation of Throughput)*. In our proposed scenario, some nodes respond to the access point by sending an RTS packet but do not communicate afterward. With simulation results, we show that this adversarial behavior of some nodes greatly impacts the throughput of the network. Our findings suggest a new research direction in protocol design in order to maintain steady throughput even in the presence of adversarial nodes.

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